Proceedings of CONF-FMCE 2025 Symposium: AI and Machine Learning Applications in Infrastructure Engineering
DOI: 10.54254/2755-2721/2025.MH25102

Chemical Stability of Cosmetic Ingredients: Mechanisms of Degradation and Influencing Factors

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Abstract. Cosmetic ingredients are crucial for the chemical stability of products, which is a key factor to judge the safety and shelf life of the product. Cosmetic ingredients are classified based on their susceptibility to these processes, with important characteristics such as molecular structure, functional groups, and reactivity influencing their stability profiles. This paper examines the significance of chemical stability in cosmetic products and how it affects their efficacy and safety. It also analyzes both internal factors, such as molecular structure and formulation composition, and external conditions, including temperature, light, and packaging. The findings reveal that both internal and external factors impact chemical stability. High temperatures and extreme pH values might accelerate degradation, whereas light can produce photochemical processes that diminish ingredient efficacy. These factors can be decreased with proper packing, such as airtight or UV-blocking containers. how antioxidants, encapsulation technology, and intelligent packaging are key components of stability-boosting techniques. The durability of components like vitamin C may be significantly increased by antioxidants like vitamin E and ferulic acid, which stop oxidative destruction. Liposomes and microencapsulation are two encapsulation techniques that shield delicate substances like retinol from air and light. It has also been demonstrated that intelligent packaging solutions, including UV-blocking materials or self-regulating packaging, improve product stability. The stability of cosmetic ingredients plays an essential role in maintaining product quality, safety, and effectiveness throughout its shelf life.

Keywords: Cosmetic stability, Degradation mechanism, Formulation strategy, Antioxidants, Encapsulation technology

1. Introduction

Early cosmetic formulations faced significant challenges with stability. For example, in the middle of the twenty-first century, products such as serums and creams are frequently damaged or separated because they lack the necessary packaging and preservation [1]. The failure to comprehend the methods of degradation caused a variety of problems, including inconsistent product performance and even safety. Since the cosmetic industry developed, so did the demand for scientific research in a strict way. Research started to concentrate on the reasons for the deterioration of ingredients and how to avoid them, which was the basis of contemporary cosmetic science.

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Current research has focused on a number of strategies to improve the stability of cosmetic ingredients. To find out how vitamin C is utilized in the body, Pinnell et al. examined how antioxidants like ferric acid and vitamin E stabilize it [2]. Their findings indicate that the combination of these antioxidants greatly lowers oxidation and increases the effectiveness of vitamin C in skincare formulations [3]. Their research found that the encapsulated retinol's capacity to increase stability in UV light resulted in longer-lasting cosmetic product activity [4]. In order to improve packaging materials, they investigated the effect of intelligent packaging solutions such as UV-blocking containers on the protection of sensitive materials [3]. Their findings reveal that advanced packaging materials effectivley lessen light-induced deterioration and prolong the shelf life of products.

This study uses a literature review method to analyse the chemical stability of cosmetic components. It systematically discusses degradation mechanisms, environmental and formulation factors influencing stability, and numerous enhancement methods, including antioxidants, encapsulation technologies, and intelligent packaging. This research gives valuable insights for cosmetic formulators aiming to create more durable and efficient products. Also, the results serve as a reference for future studies attempting to increase formulation procedures and sustainable packaging solutions in the cosmetic business.

2. Chemical stability of cosmetic ingredients

The chemical stability of cosmetic ingredients is mainly regulated by four primary degradation pathways, each involving different reaction mechanisms: oxidative degradation, hydrolysis, photodegradation, and thermal degradation.

2.1. Oxidative degradation

Oxidative degradation is a critical process that begins with the formation of radicals, particularly in unsaturated molecules. Allylic sites are especially susceptible to attack by molecular oxygen, leading to chain reactions that compromise ingredient integrity. In addition to unsaturated molecules, proteins and amide functionalities are also affected. For instance, vitamin C (L-ascorbic acid), a widely used active ingredient, rapidly oxidizes when exposed to air, turning yellow and losing efficacy unless stabilized properly [5]. This degradation follows a free radical mechanism involving the enediol structure of vitamin C.

To protect oxidation-sensitive actives like retinoids, various strategies are employed. These include the use of radical quenchers like vitamin E and ferulic acid, oxygen-impermeable packaging, or anhydrous delivery systems.

2.2. Hydrolytic degradation

Hydrolysis primarily affects ester and amide bonds, with degradation rates strongly influenced by pH. Polysorbates, which are common non-ionic surfactants, undergo hydrolysis of their ester bonds in aqueous systems, particularly at extreme pH values [6]. This process compromises emulsion stability. While parabens remain relatively stable, other natural preservatives like essential oils can show inconsistent antibacterial efficacy unless carefully formulated [7].

Stabilization of hydrolytically sensitive components often involves the use of pH buffers and moisture barrier technologies.

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2.3. Photodegradation

Photochemical instability arises when UV light is absorbed by chromophores in a molecule, causing electronic transitions that lead to isomerization or bond dissociation. Retinol derivatives, in particular, are highly photosensitive and can degrade rapidly when exposed to UV light. Studies have shown that these compounds can experience nearly complete degradation within just 8 hours of UV exposure if not adequately protected [8].

To safeguard photolabile ingredients, formulators can utilize opaque packaging that prevents light penetration and incorporate UV-absorbing excipients that shield sensitive molecules from harmful radiation.

2.4. Thermal degradation

Thermal degradation follows Arrhenius behaviour, meaning that as temperatures rise, the increased energy can facilitate bond cleavage and molecular rearrangement. Thermally sensitive compounds benefit from phase-stabilizing matrices designed to buffer against temperature changes.

Thermal degradation follows Arrhenius kinetics, where elevated temperatures provide sufficient activation energy to break chemical bonds, leading to molecular rearrangement or decomposition. This process is particularly critical for heat-sensitive compounds such as peptides, enzymes, and certain antioxidants (e.g., coenzyme Q10). For instance, studies show that coenzyme Q10 loses over 50% of its activity after exposure to temperatures above 40°C for 30 days due to the cleavage of its quinone ring structure [8].

2.5. Structural considerations and integrated stabilization

The chemical structure and classification of ingredients significantly influence stability. Rational stabilization requires a combination of physical barriers (e.g., packaging) and molecular-level treatments (e.g., chelation of pro-oxidant metals, steric hindrance of reactive sites). Advanced analytical tools such as electron paramagnetic resonance and high-resolution mass spectrometry enable accurate mapping of degradation pathways.

Recent advancements in encapsulation—such as the use of silica or cyclodextrin matrices—have shown promise in protecting unstable actives while preserving their bioavailability [9]. A thorough evaluation of the entire formulation is essential, as ingredient interactions may lead to unexpected degradation risks.

3. Factors affecting the chemical stability of cosmetic ingredients

The stability of cosmetic components is influenced by a range of external and internal factors. These can be broadly divided into environmental variables and formulation interactions.

3.1. Environmental variables

Environmental conditions play a significant role in determining the degradation rate of cosmetic ingredients. Temperature fluctuations are a major destabilizing factor. Elevated temperatures increase molecular mobility and accelerate reaction rates, thereby enhancing the kinetics of degradation.

pH extremes similarly influence ingredient stability. For example, niacinamide hydrolyses when the pH falls outside its optimal range of five to seven, leading to reduced efficacy [10].

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Light exposure is another critical concern. Transparent packaging allows UV radiation to penetrate the formulation, triggering photochemical reactions in sensitive compounds. This is particularly problematic for ingredients like vitamin C derivatives and retinoids. To address this, light-resistant or UV-blocking containers are often used to reduce light-induced degradation and prolong product shelf life.

3.2. Formulation interactions

Formulation interactions play a pivotal role in determining the chemical stability of cosmetic ingredients. The compatibility between active compounds, excipients, and the base matrix can either enhance or compromise product integrity. For example, cationic surfactants (e.g., cetrimonium bromide) may destabilize anionic actives (e.g., hyaluronic acid) through electrostatic interactions, leading to precipitation or reduced efficacy [10]. Similarly, the coexistence of antioxidants (e.g., vitamin E) and metal ions (e.g., iron) in emulsions can accelerate oxidative degradation by catalyzing free radical formation, even in trace amounts [8].

Key interaction mechanisms in formulation stability include solubility effects, pH-dependent reactions, and competitive oxidation. Poorly soluble actives, such as flavonoids, can crystallize over time in aqueous formulations, which alters their bioavailability. To mitigate this issue, solubilizers like cyclodextrins or ethoxylated emulsifiers can be employed to form inclusion complexes or micelles that enhance solubility. Additionally, pH-dependent reactions pose a risk, as combining ingredients with incompatible pH ranges can trigger degradation. For example, mixing acidic vitamin C (pH ~3.5) with alkaline peptides (pH ~7) can lead to hydrolysis of both compounds unless dual-chamber packaging is used to isolate them. Furthermore, competitive oxidation is a concern, particularly with polyunsaturated lipids like squalene, which compete with antioxidants for oxygen. This competition can deplete stabilizers and shorten the shelf life of formulations, necessitating either higher antioxidant concentrations or the use of oxygen-scavenging packaging to maintain product integrity. 4.Stability Enhancement Strategies

Maintaining the stability of active cosmetic components is crucial for ensuring product efficacy and extending shelf life. Several well-established strategies have been developed to counteract specific degradation mechanisms.

4. Stability enhancement strategies

4.1. Antioxidant systems

Synergistic antioxidant combinations have proven particularly effective for stabilizing oxidation-prone substances such as vitamin C (ascorbic acid). Pinnell et al. demonstrated that the combined use of vitamin C, ferulic acid, and vitamin E (α-tocopherol) significantly slows oxidative degradation and enhances photoprotection [2]. Ferulic acid stabilizes the intermediate radical form of vitamin C, while vitamin E terminates the oxidation chain reaction. Together, these antioxidants function as a complementary radical-scavenging system.

4.2. Photostability enhancements

Photolabile compounds like retinol present unique challenges. Barraza et al. reported that liposomal encapsulation offers substantial protection against UV-induced degradation [3]. Liposomes, composed of phospholipid bilayers, serve as physical barriers that absorb harmful UV light while

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maintaining the bioavailability of retinol. This technique has been successfully integrated into various anti-aging formulations.

Additional photoprotection can be achieved using UV-blocking agents. Soni et al. found that the inclusion of UV absorbers in retinoid formulations led to a 70% reduction in photodegradation and significantly extended product shelf life [11]. These findings have led to the development of combined formulations containing both retinoids and physical sunscreens such as zinc oxide.

4.3. Encapsulation technologies

Encapsulation methods extend beyond retinoids to include other sensitive actives. Salvi et al. explored microencapsulation within polymer matrices, which protects against oxidative and hydrolytic degradation while enabling controlled release [4]. Encapsulation using silica or cyclodextrin carriers has shown effectiveness in stabilizing heat-sensitive compounds and maintaining their performance in modern formulations. For instance, in the cosmetics industry, encapsulated active ingredients can be released gradually over time, providing a sustained effect and enhancing the overall efficacy of the product.

4.4. pH optimization and packaging innovations

pH control is another key stabilization strategy. McLafferty highlighted that vitamin C remains most stable at pH values below 3.5 [12]. In contrast, peptides are best preserved in near-neutral conditions to avoid hydrolysis. Based on such findings, dual-chamber packaging systems have been developed to separate incompatible actives until the point of application, thereby preserving their stability and effectiveness. These advanced packaging solutions not only enhance the shelf - life of products but also ensure that the actives retain their potency when used, making them highly beneficial for formulations containing multiple delicate ingredients.

5. Conclusion

The chemical stability of cosmetic ingredients is important to ensure that products are both effective and safe for consumers. Cosmetic formulators may create products that maintain their integrity over time by understanding the various degradation mechanisms (oxidation, hydrolysis, and photodegradation) and the variables affecting these processes (temperature, pH, and packaging). Stability-enhancing strategies, like antioxidants, encapsulation, and intelligent packaging, play a significant function in protecting active compounds and safeguarding their efficacy. Using active compounds like antioxidants (like vitamin E and ferulic acid) and stabilizing techniques (such encapsulation and UV-blocking agents) can significantly extend the shelf life of cosmetically prepared items. The stability of cosmetics is greatly influenced by formulation and environmental factors, including temperature, light, and pH. Efficiency The development of improved methods has been successful in decreasing the amount of ingredient degradation, increasing the overall quality of the finished product. Methods such as encapsulation, intelligent packaging, and the application of stabilizers all help reduce the hazards related to ingredient breakdown.

While this study provides an overview of the degradation mechanisms and stabilization strategies for cosmetic ingredients, it is not without limitations. First, the discussion is primarily based on existing literature, and lacks experimental validation or quantitative comparison of the effectiveness of different stabilization methods. Furthermore, most examples focus on widely studied actives like vitamin C and retinol, whereas emerging natural or biotechnologically derived ingredients receive

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comparatively little attention. Incorporating more data on newer or less conventional actives would enrich the analysis.

Looking ahead, future research can focus on several promising directions. One area is the development of biodegradable and sustainable packaging materials that not only preserve ingredient stability but also reduce environmental impact. The development of multifunctional excipients that provide pH buffering, photoprotection, and antioxidant activity all at once is another crucial avenue. Additionally, the integration of real-time analytical tools into product development could allow formulators to monitor stability under dynamic storage conditions. These advancements would support the creation of safer, longer lasting, and more environmentally conscious cosmetic products.

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